

SEID 2364 – Assignment 5 System Structuring and Ethical Analysis

Part 1 – System Structure (Descriptive Layer)

Decision Statement

An AI hiring system rejects a job applicant based on automated resume screening.

System Components

Actors

The primary actors in this system include the company using the AI hiring tool, human recruiters, software developers, and the AI system itself. Recruiters interact with the recommendations produced by the system, while developers design and maintain the algorithms used for screening applicants.

Targets

The primary targets are job applicants because the system directly evaluates and ranks them. Secondary targets include families or dependents of applicants because employment outcomes can affect financial stability and quality of life.

Determinants

The main determinants are company hiring policies, algorithmic rules, training datasets, and qualification criteria. These factors strongly shape which applicants are considered suitable for employment.

Contributors

Contributors include HR staff, previous hiring records, third-party software vendors, and external datasets used for training the AI system. Although they may not make the final decision directly, they influence how the system operates.

Information and Data Inputs

The system uses resumes, educational records, work experience, skill assessments, and application forms as inputs.

Source: The information comes mainly from applicants, company records, and external databases.

Authority: The company and software providers control how the information is stored, processed, and evaluated.

Trust Basis: The information is trusted because applicants are expected to provide accurate details, while companies rely on verification processes and historical records.

Contrasting Example

If two applicants apply for the same role, the system may produce different outcomes depending on the data available. An applicant with work experience matching historical hiring patterns may be ranked highly, while another equally capable applicant with a non-traditional background may be rejected. This shows how the same system can generate different outcomes based on changing inputs and conditions.

Part 2 – Interpretive Layer (Ethical Overlay)

Authority within the system is mainly held by the company and the AI developers because they define hiring criteria and system rules. Ownership of data and algorithms is controlled by the company and software providers. Responsibility, however, becomes distributed because recruiters may blame the AI system while developers may argue that the company controls deployment decisions.

Agency exists in both humans and technical systems. Recruiters have the ability to approve or reject recommendations, while the AI system influences outcomes through ranking and filtering applicants.

Harm occurs when biased or inaccurate recommendations unfairly affect applicants. Economic harm may result from lost employment opportunities, while social harm may appear through discrimination against specific groups.

Part 3 – System Flow (Pre-Framework)

1. Applicants submit resumes and personal information. 2. The AI system processes and analyzes application data. 3. Machine learning models compare applicants against hiring criteria. 4. The system ranks or filters candidates. 5. Recruiters review recommendations generated by the system. 6. Hiring decisions are made and communicated to applicants.

Throughout this process, information flows between applicants, databases, algorithms, and recruiters. Decisions are not located in a single point because both human judgment and automated systems contribute to the final outcome.